M. Vidhya

192412151

Experiment 9: Compare Linear and Polynomial Regression

Aim

To implement and compare **Linear Regression** and **Polynomial Regression** using Python.

Python Program

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error, r2_score

# Dataset
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([2, 4, 5, 4, 5, 7, 9, 10, 12, 15])

# Linear Regression
linear_model = LinearRegression()
linear_model.fit(X, y)
linear_pred = linear_model.predict(X)

# Polynomial Regression
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)
poly_pred = poly_model.predict(X_poly)

# Evaluation
print("Linear Regression")
print("Mean Squared Error:", mean_squared_error(y, linear_pred))
print("R2 Score:", r2_score(y, linear_pred))
```

```
print("\nPolynomial Regression")
print("Mean Squared Error:", mean_squared_error(y, poly_pred))
print("R2 Score:", r2_score(y, poly_pred))

# Plotting
plt.scatter(X, y, label="Actual Data")
plt.plot(X, linear_pred, label="Linear Regression")
plt.plot(X, poly_pred, label="Polynomial Regression")
plt.xlabel("X values")
plt.ylabel("Y values")
plt.title("Linear vs Polynomial Regression")
plt.legend()
plt.show()
```

Sample Output

Linear Regression

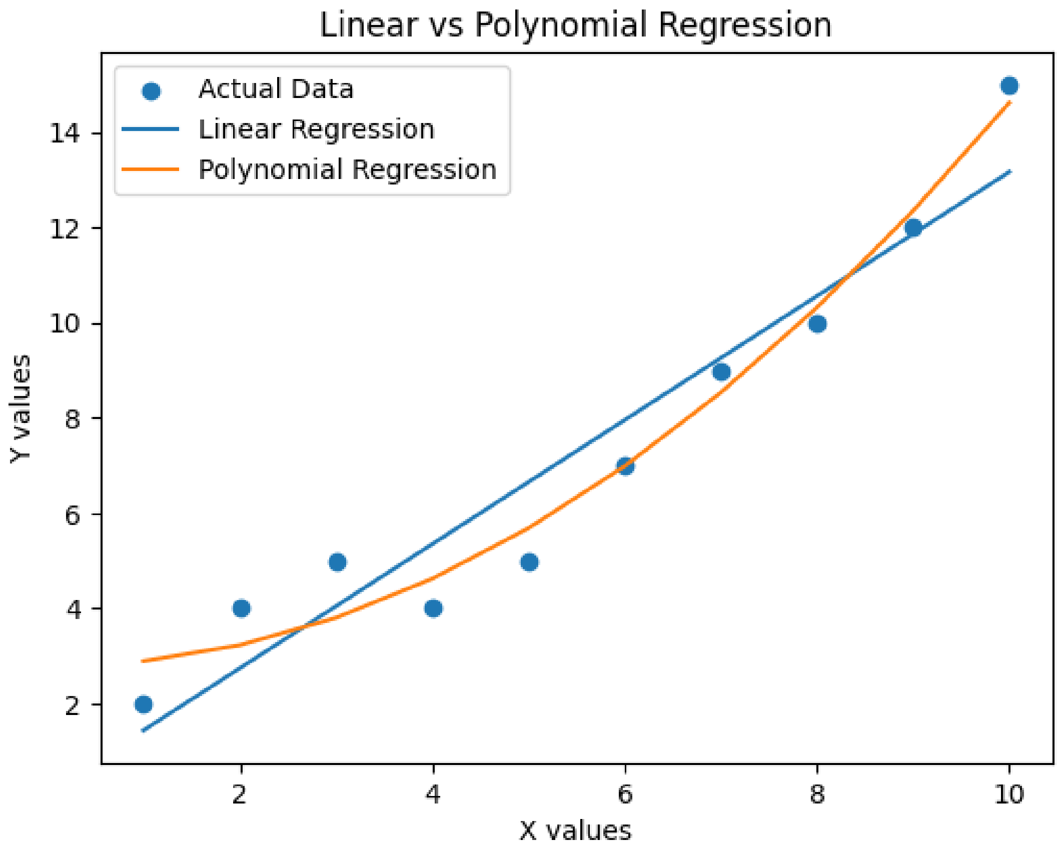
Mean Squared Error: 1.2024242424242426

R2 Score: 0.9209451517143825

Polynomial Regression

Mean Squared Error: 0.42666666666666647

R2 Score: 0.9719482796405874


Linear vs Polynomial Regression
Actual Data
Linear Regression
Polynomial Regression
Y values
X values
2
4
6
8
10
12
14
2
4
6
8
10